

PSK guessing: As we learnt earlier, a pre-shared key is used between the wireless AP and node to encrypt communication. Typically, administrators setting up Wi-Fi networks tend to leave the vendor-provided default key in place. Smart attackers usually first try to detect the manufacturer of wireless APs, and if that information is not available, they try to guess it and attempt to break the key.

Key cracking: Usually a pre-shared key should be enough to establish security. However, in case of WEP-based Wi-Fi networks, attackers can use passive methods to sniff and capture a lot of data, and subject it to key-cracking algorithms. As we saw earlier, WEP is a simple RC4 XOR type of encryption, and it only takes some amount of time to break into it. It had been demonstrated by attackers that a packet capture of more than 40000 can be sufficient data to crack a WEP key in minutes. With the introduction of WPA security features in a wireless AP, it became tough to break the key. However other brute-force attacks, such as statistical key guessing, dictionary attacks, etc, can be used to crack it.

Wireless attack detection

Before we talk about protecting the infrastructure, it is imperative to understand a few detection techniques. Unlike wired networks, a wireless network signal can be compromised easily, which makes detection difficult but certainly not impossible.

AP monitoring: As we learnt, securing the SSID of an AP or wireless router is very important. In a large organisation, keeping track of SSIDs can be a challenge; hence, this information should be programmatically stored in a secure database. Other crucial details, such as the MAC ID, IP restrictions, the wireless channel used, the beacon settings, wireless signal strength and bandwidth type are stored for each corresponding SSID. A wireless monitoring device, or a mobile device running monitoring software, is used to detect all stations and APs periodically, and the results are compared with the baseline database created earlier. Such routine audits ensure the integrity of router settings and thus the overall wireless network security.

Wi-Fi node monitoring: Along with the APs, each node needs to be monitored too. The technique is a bit different, though. For the nodes, a MAC-based security on the APs can be configured, whereby a particular AP would support only a set of MAC addresses. This ensures that the wireless client node cannot roam around beyond the configured zone, and if such a need arises the request can be fulfilled via an authorisation and approval process. For large organisations, this can result in systems administration overhead, in which case the nodes can be allowed to connect to all APs; however, each connection and disconnection can be logged and parsed for anomalous behaviour.

Traffic monitoring: Besides the above techniques, network administrators can periodically take samples of data from each AP, and check for denial of service and SYN

flood attacks. Multiple connections and disconnections on a particular AP from one or more client nodes should also trigger a warning. As for Layer 2 attacks, a signal spectrum detection tool can be incorporated too, to detect signal-jamming situations.

Protecting FOSS systems

Along with the monitoring techniques, additional security measures are essential. For small networks, changing the default password and SSID of the AP is a must. Modern routers are equipped with a feature to disable the broadcasting of SSID, which should be turned on to ensure that passive sniffing attacks are thwarted to some extent. Periodically changing SSIDs is highly recommended, though it can be a tough task for a large number of wireless APs. To protect a Linux server farm hosted in a data-centre, the wireless signal strength of APs should be adjusted in such a way that it should be adequate for client nodes to connect and transfer data seamlessly, but at the same time it should not cross physical building boundaries, whereby it can be detected by a 'drive-by' attacker.

Using WPA2 security instead of WEP is recommended. Besides, the shared key of WPA security should be long and complex enough to stop directory brute-force attacks. For large corporations, the Layer-7 wireless security software should be installed on client nodes as well as APs, to further strengthen the encryption process. For FOSS systems, using an X.509 certificate on either end of the wireless communication can help achieve cheaper yet effective security. Most famous flavours, such as Debian and Ubuntu, support WPA2 security with trimmings such as AES, TKIP and LEAP. Configuring those, along with MAC address filtering, and enabling firewall features can protect a serious server farm, yet let it enjoy the benefits of wireless networks.

Wireless attacks are, unfortunately, easy to carry out and difficult to detect. Modern data centres allow the presence of wireless networks connected to the product server farms, thus requiring the implementation of security measures. While there is no single solution to protect wireless networks, an appropriate combination of the techniques mentioned above can achieve adequate security. Wireless monitoring audits are an important activity that needs to be done by network administrators at regular intervals. 

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